

AZPIPE

Fold-locked API path for aziel-runtime

AP-WP-0.2 9 September 2026 Author: Aziel Eliab / GodLock.AZ Status: local specification. Not a fleet-completeness claim. Runtime: `azpipe.py` · Magic: FLD3 · Version: AZPIPE-0.2

0. Sentence

All data on the API, in and out, travels a fixed hop list. Architecture is visible on the envelope. Memory and tools never see raw inbound bytes.

1. Claim

A Worker, a library call, a mesh cite, and an MCP tool response are the same class of object once they enter aziel-runtime: a payload that must be refused or admitted. AZPIPE is the admission path.

FoldLock is the suppression hop. StaticClock-style freeze is the hash hop. FragGate is the refuse/door hop. SweepGate is the airlock. ChainLock is entry. Toolkits run last.

The hop list is the architecture. Changing product names does not change the list.

2. What it is not

- Not zip and not encryption. FoldLock is tether-word suppression on UTF-8.
- Not a TETH-1 completeness claim while `foldlock.py` is absent. Current hop is `fld3-wire`: allowlisted origins plus block-keys.
- Not a public Node Gate panel.
- Not QNM launch. The envelope reserves two bridges, isolate, and no body pull. Cells of 25 are specified elsewhere. This pipe does not join a cell.
- Not a substitute for host AV, sandboxing, or EmbryoLock wipe.

3. Hops

Inbound:

`frag → sweep → fold → static → fold → entry → frag → toolkits`

Outbound:

`toolkits → frag → fold → static → fold → sweep → frag`

frag (first). DecisionGATE flags D/E/I/G/R. Empty or ungrounded inbound refuses before SweepGate spends work. FragGate remains the kernel. This hop does not replace FragGate.

sweep. SweepGate airlock. Poison marks, malware-class structural marks, block-keys, off-origin URLs. Isolate and refuse. Do not merge. See SG-WP-0.1.

fold (first). FoldLock FLD3. If `foldlock.py` sits beside `azpipe.py`, load `suppress / fold / tether_suppress`. Otherwise `fld3-wire`: non-allowlisted `http(s)` becomes `[FLD3:url]`; keys named `password`, `private_key`, `secret`,

ssn, legal_name, home_address become [FLD3:block].

Allowlist: <https://www.azielcorpuslibrary.net>, <https://godlock.uk>, <https://www.azieleliab.com>, <https://aziel-runtime.vibelock.workers.dev>.

static. Canonical JSON (sorted keys, tight separators). Body SHA-256. Fact SHA-256. Pinned UTC timestamp. The freeze is what later hops hash. Volatile paths do not belong in the frozen body.

fold (second). Defense in depth. Tethers that appear only after canonicalization get the same suppression.

entry. ChainLock. Stamp, interact, recall, seal. The payload is now a fact-bearing card.

frag (second). Door before toolkits. Grounded claim or refuse. Toolkits do not run on an ungrounded envelope.

toolkits. Named MCP tools only.

Outbound is the reverse so a dirty toolkit result is swept before it rides a QNM bridge.

4. Envelope

```
{
  magic: "FLD3",
  v: "AZPIPE-0.2",
  dir: "in" | "out",
  path: [...hops],
  ok: true | false,
  h: sha256(inner),
  fh: sha256(clipped fact),
  isolate: {in, out},
  gates: {D,E,I,G,R,refuse},
  sweep: {airlock, hits, isolate, pull},
  foldlock: bool,
  teth: "foldlock.py" | "fld3-wire",
  qnm: {ready, pull:false, bridges:2, isolate:true},
  inner: {entry, toolkit, static}
}
```

Refuse envelopes stop at the hop that closed. They do not carry inner into memory.

Every ChainLock emit card also carries a thin pipe:{v,magic,path,h,fh,iso,qnm} so MCP output stays under 4096 bytes without dumping the full envelope. Full envelope: `chainlock.py pipe --dir in|out --json '{...}'`. Architecture: `pipe --arch`.

5. Isolate

Both directions isolate on the fact hash fh, not the whole stamp dump. Body hash remains on the freeze for byte integrity. QNM cites fh. Remote body does not enter STM.

Two payloads with the same fact text share fh. Two payloads with the same fact text and different wrapping URLs do not share body hash after fold, because off-origin URLs fold to a mark. That is intended.

6. Why this order

Frag first so SweepGate does not chew empty noise.

Sweep before fold so poison never becomes a folded "clean" string.

Fold before static so the freeze hashes the folded bytes.

Second fold after static so canonicalization cannot reintroduce a tether.

Entry only after the airlock is open.

Second frag so toolkits cannot run on an ungrounded card.

Outbound sweep so the mesh never ingests a toolkit leak.

A shorter list (frag–sweep–entry–frag) drops FoldLock and the freeze. That is cheaper and weaker. Long-term QNM needs both.

7. QNM slot

bridges=2, pull=false, isolate=true, sweep-before-ingest. Poison drops the tether. Tamper isolates from chain. This paper does not claim a live fleet or a qubit.

8. Cap

f1d3-wire is not TETH-1. Drop `foldlock.py` next to `azpipe.py` to switch lexicons without changing hops.

Companion papers: CL-WP-0.4, SG-WP-0.1, LS-WP-0.1.